

Problem Statement : Banking Simulation – C Programming Project

GROUP 15

FOP (FOUNDATION OF PROGRAMMING)

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BANK ACCOUNT SIMULATION USING C

Designing a **C program** to simulate fundamental banking operations.

- **Core Banking Functions**

Users can easily deposit, withdraw, and check their balance.

- **System Navigation**

The program provides a clear option to exit the system gracefully.

PROJECT OBJECTIVES

- 1 Practical Implementation**
Understand real-world application of C programming concepts.
- 2 Simulate Banking Operations**
Mimic basic financial transactions in a controlled environment.
- 3 Conditional Statements**
Mastering if-else logic for dynamic program flow.
- 4 Looping Concepts**
Apply iterative structures for repeated actions.
- 5 Logical Thinking**
Enhance problem-solving skills through structured design.

KEY 'C' CONCEPTS UTILISED



Variables

Storing dynamic data within the program.



Data Types

Using int and float for numerical values.



Arithmetic Operators

Performing calculations like addition and subtraction.



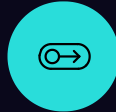
Relational Operators

Making comparisons between values.



Conditional Statements

Implementing if-else for decision-making.



Switch Case

Handling multiple choices efficiently.



Loops

Utilising while loops for repetitive tasks.

AIM OF THE PROJECT

Learn C fundamentals

Variables, control flow, input/output 4
core building blocks

Boost problem-solving

Design decisions, stepwise logic, and
testing



Apply logic to real life

Translate everyday banking tasks
into
program steps

Simulate basic operations

Deposit, Withdraw, Balance enquiry.

ADVANTAGES OF THE PROJECT



Easy to implement

Minimal syntax, no advanced features — ideal for beginners

Strengthens logic

Branching and loops to handle deposit/withdraw flows

1

2

3

4

Real-world relevance

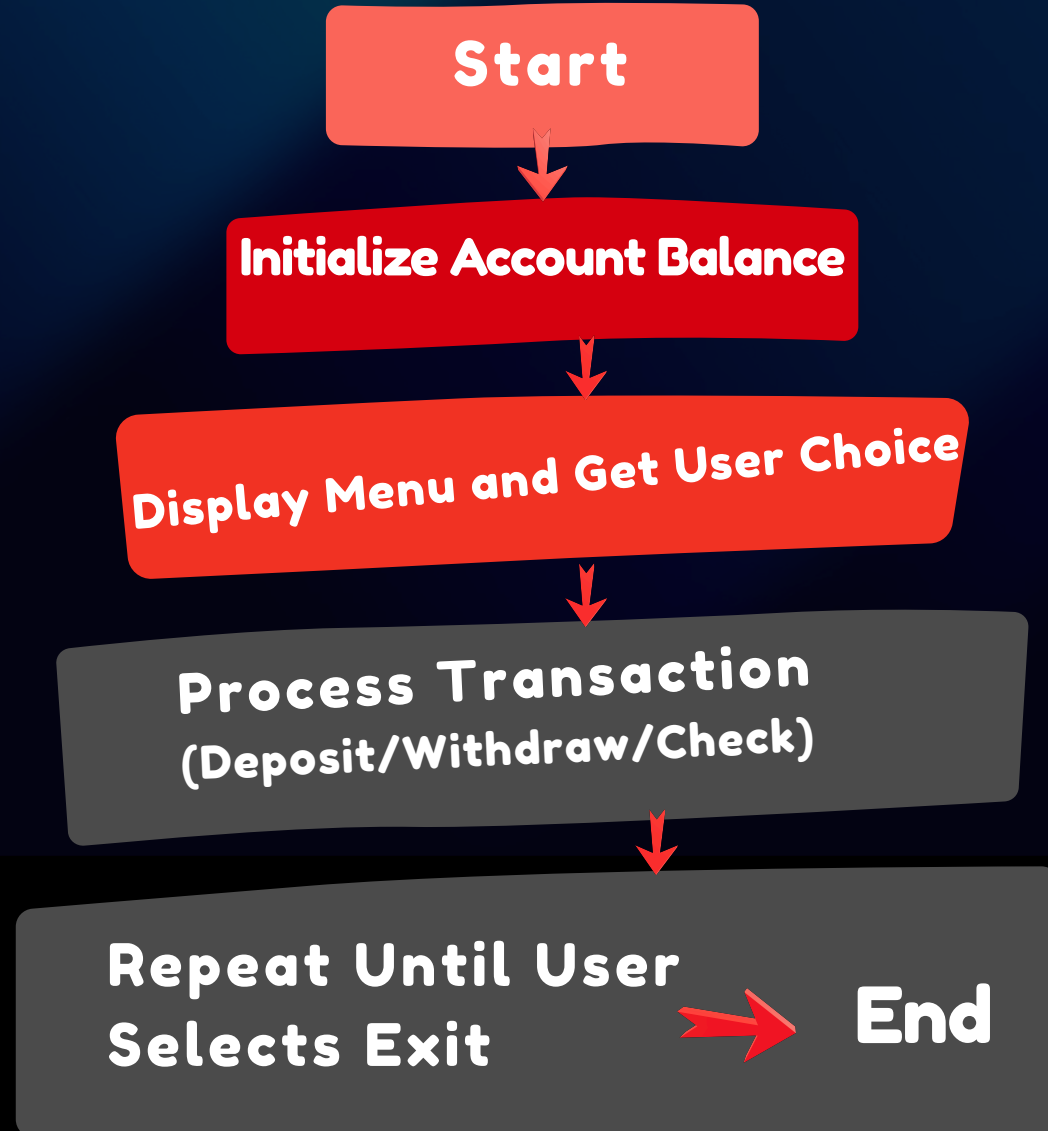
Show show programs model everyday tasks like banking

Solid foundation

Prepares students for arrays and functions later

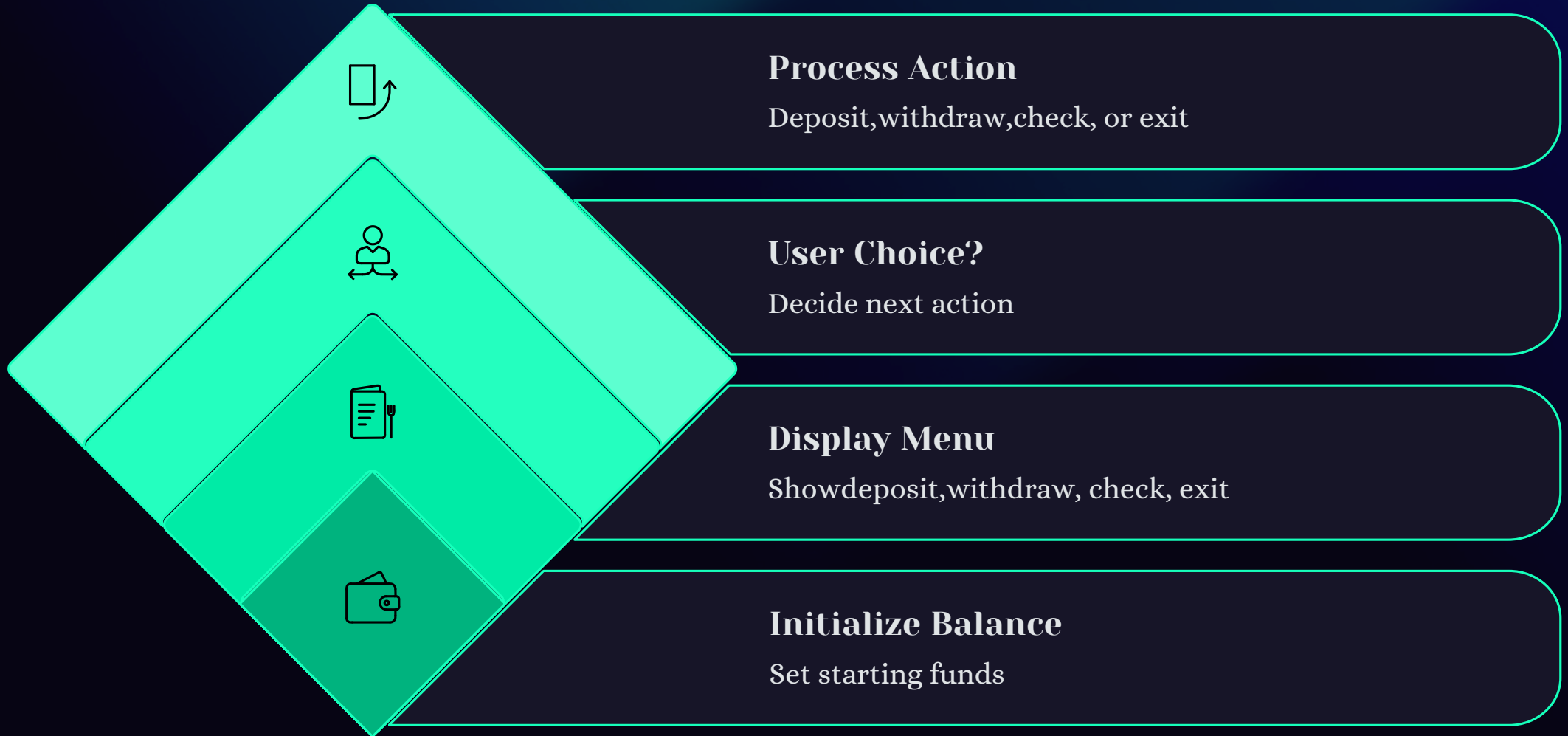
PROGRAM ALGORITHM: STEP-BY-STEP

“A logical flow to guide the banking simulation.”



PROGRAM FLOWCHART

"Visualising the journey through our bank simulation".



BANK SIMULATION: C CODE

```
1  #include<stdio.h>
2  |
3  void viewBalance(double balance){
4      printf("your current balance is : $%.2lf\n",balance);
5  }
6
7  void depositFunds(double *balance){
8      double deposit;
9      printf("Enter amount to deposit:$");
10     scanf("%lf",&deposit);
11
12     if (deposit > 0) {
13         *balance += deposit;
14         printf("Deposited $%.2lf. Your new balance is: $%.2lf\n",deposit,*balance);
15     }else{
16         printf("Invalid deposit amount. please try again.\n");
17     }
18
19 }
20
21 void withdrawFunds(double *balance){
22     double withdrawl;
23     printf("Enter amount to withdraw : $");
24     scanf("%lf",&withdrawl);
25     if ( withdrawl <=0 ){
26         printf("Invalid deposit amount. please try again.\n");
27     }else if( withdrawl > *balance ){
28         printf("Insufficient balance ! your balance is $%.2lf\n", *balance);
29     }else{
30         *balance -= withdrawl;
31         printf("withdrawn $%.2lf. your new balance is : $%.2lf\n",withdrawl, *balance);
32     }
33 }
34 int main(){
35     double balance = 0.0;
36     unsigned short choice;
```

```
37     do{
38         printf("\nWelcome to the Bank!\n");
39         printf("1. View Balance\n");
40         printf("2. Deposit Funds\n");
41         printf("3. Withdraw Funds\n");
42         printf("4. Exit\n");
43         printf("Enter your choice:");
44         scanf("%hu",&choice);
45
46         switch(choice){
47             case 1: {
48                 viewBalance(balance);
49                 break;
50             }
51             case 2 :{
52                 depositFunds(&balance);
53                 break;
54             }
55             case 3:{
56                 withdrawFunds(&balance);
57                 break;
58             }
59             case 4:{
60                 printf("thank you for visiting have a Good Day !!");
61                 break;
62             }
63             default:{
64                 printf("Invalid Choice.Please select a valid option.\n");
65             }
66         }
67     }while(choice !=4);
68
69     return 0;
70
71 }
```

Sample Output & Conclusion

“Sample Interaction”

“This interaction demonstrates a successful deposit, reflecting the program's functionality.”

Conclusion :

→ Successfully implemented bank account simulation in C.

→ Applied fundamental C programming concepts effectively.

→ Enhanced logical thinking through loops and conditional structures.



OUTPUT :

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:4
```

```
thank you for visiting have a Good Day !!
```

```
=== Code Execution Successful ===
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:2
```

```
Enter amount to deposit:$100
```

```
Deposited $100.00. Your new balance is: $100.00
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:|
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:1
```

```
your current balance is : $0.00
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:3
```

```
Enter amount to withdraw : $100
```

```
Insufficient balance ! your balance is $0.00
```

```
Welcome to the Bank!
```

1. View Balance
2. Deposit Funds
3. Withdraw Funds
4. Exit

```
Enter your choice:
```

PROJECT

BANK ACCOUNT SIMULATION USING C

TEAM MEMBERS - GROUP 15:

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PROBLEM STATEMENT & OVERVIEW

Goal: Design a C program for fundamental banking operations.

Core Functions: Deposit, Withdraw, Check Balance, Exit gracefully.

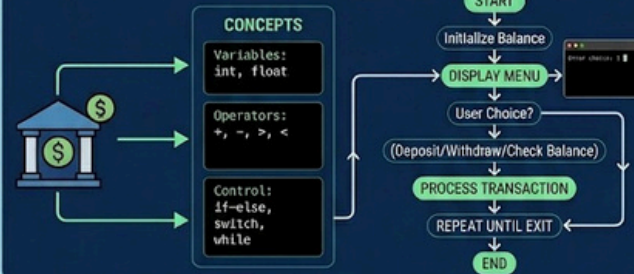
Purpose: Practical application of C concepts.

4. SAMPLE INTERACTION (DEPOSIT)

```
Enter amount to deposit: 500
Deposit successful.
New balance: 500.00
```



3. KEY 'C' CONCEPTS UTILIZED & PROGRAM FLOW



2. PROJECT OBJECTIVES



- Implement real-world C programming.
- Simulate financial transactions.
- Master logic (If-else, loops).
- Enhance problem-solving skills.



5. CONCLUSION & ADVANTAGES

- Successfully implemented simulation.
- Easy to implement with minimal syntax.
- Strengthens logical thinking.
- Real-world relevance (banking).
- Solid foundation for advanced topics.



Thank You !!
